

Sensor Fusion SLAM on the SubSurfaceGeoRobo Underground Dataset

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Abstract—Subterranean environments present significant challenges for Simultaneous Localization and Mapping (SLAM), including narrow corridors, repetitive geometry, airborne particulates, and degraded sensing conditions that can reduce registration accuracy and increase drift. While LiDAR provides accurate geometric measurements, its performance may deteriorate in perceptually degraded tunnel environments, motivating the use of complementary sensing modalities. In this work, we present a multi-sensor SLAM study on the SubSurfaceGeoRobo dataset, a real underground mine benchmark containing synchronized LiDAR, millimeter-wave radar, wheel odometry, and high-precision RIEGL ground truth. We develop a custom LiDAR-odometry pipeline using scan-to-submap registration and local mapping, and extend it with radar-based motion constraints for improved robustness. We further evaluate both systems under simulated underground LiDAR degradation. Results show that radar fusion reduces mean map error by 36% under standard degradation and 61% under heavy degradation, recovering near-baseline accuracy while introducing negligible impact under clean conditions, demonstrating the practical benefit of radar integration in challenging subterranean environments.

Index Terms—SLAM, LiDAR odometry, millimeter-wave radar, sensor fusion, subterranean environments, geomonitoring

I. INTRODUCTION

Simultaneous Localization and Mapping (SLAM) lets autonomous robots estimate their trajectory and build a map of their surroundings in GPS-denied settings, supporting tasks like navigation, inspection, and geomonitoring [1]. LiDAR-based SLAM is the dominant approach, combining accurate 3-D point clouds with short-term motion estimates from sensors such as IMUs or wheel encoders [2]. Underground environments, however, break SLAM in ways that standard benchmarks rarely capture: long featureless corridors make scan-matching ambiguous along the tunnel axis; repeated rock textures make loop closures hard to detect; airborne dust and water spray corrupt LiDAR returns; and wet floors reflect the laser, adding false range measurements [3].

These challenges are highly relevant for real-world mine surveying, tunnel inspection, and infrastructure monitoring, where GPS is unavailable and repeated physical access is costly. Despite rapid progress driven by competitions such as the DARPA Subterranean Challenge [4], state-of-the-art

LiDAR-inertial frameworks can still diverge in extremely narrow or featureless underground corridors.

Millimeter-wave (mmWave) radar operates at 77–82 GHz and is fundamentally unaffected by optical occlusion, dust, smoke, or humidity—precisely the conditions that degrade LiDAR [5]. Radar-aided odometry and SLAM have been explored for adverse outdoor environments, but integration in tightly coupled underground SLAM frameworks remains underexplored, and no prior study benchmarks LiDAR-radar fusion on a public underground dataset with sub-centimeter ground truth.

This paper makes three contributions:

- 1) We develop a custom LiDAR-odometry SLAM baseline tailored to narrow underground tunnel environments.
- 2) We integrate millimeter-wave radar as an auxiliary motion constraint and compare LiDAR-only, radar-only, and fused configurations.
- 3) We evaluate robustness under simulated LiDAR degradation using trajectory and mapping metrics against ground truth.

II. RELATED WORK

A. LiDAR Odometry and SLAM

LiDAR odometry via iterative point cloud registration has been foundational to autonomous navigation in GPS-denied environments. LOAM [8] introduced feature-based scan matching using edge and planar points to achieve real-time odometry with low drift, and remains a widely used baseline. Tightly coupled LiDAR-inertial extensions such as LIO-SAM [9] and FAST-LIO2 [10] improve robustness through IMU preintegration. In this work, we use a lightweight LiDAR-odometry design with wheel-encoder motion priors instead of IMU preintegration, trading tight inertial coupling for simplicity and avoiding known noise issues in the dataset’s IMU.

B. Subterranean SLAM

The DARPA Subterranean Challenge [4] drove significant progress in multi-sensor fusion for underground autonomy. LAMP 2.0 [1] demonstrated robust multi-robot SLAM in mine, tunnel, and cave environments via aggressive outlier

TABLE I: Comparison of Underground SLAM Datasets

Dataset	LiDAR	Radar	IMU	GT	Min. W
SubT-MRS [3]	✓	–	✓	TLS+TS	–
MIN3D [3]	✓	–	✓	TLS	5 m
SubSurfGeoRobo [6]	✓	✓	✓	TLS+TS	0.95 m

rejection and distributed pose graph optimization. Ebadi et al. [3] survey SLAM systems tested in the SubT Challenge and identify corridor degeneracy and illumination failure as the primary open problems remaining after the competition. The SubSurfaceGeoRobo dataset [6] provides the most comprehensive underground benchmark available, with synchronized multi-modal sensing and a sub-centimeter terrestrial laser scanner (TLS) reference, enabling rigorous evaluation of SLAM algorithms under real mine conditions.

C. Radar-Aided SLAM

Millimeter-wave radar provides measurements robust to dust, smoke, and lighting changes. Fritsche et al. [11] fuse LiDAR and radar for SLAM in harsh outdoor environments, showing that radar measurements anchor odometry when LiDAR returns are degraded by precipitation or fog. Radar-inertial odometry [12] exploits the Doppler ego-velocity measurement of 4-D mmWave radars to constrain IMU integration. Most radar SLAM work targets adverse outdoor conditions; integration in underground SLAM with high-precision ground truth evaluation has not been previously reported.

III. DATASET

The SubSurfaceGeoRobo dataset [6] was collected in the Reiche Zeche Research Mine in Freiberg, Germany, 150 m underground at a constant 10–11°C. The sensor platform is a four-wheel-drive UGV (Robotnik) carrying: a Robosense RS-LiDAR-16 3-D LiDAR (360° horizontal / 30° vertical, ± 2 cm ranging accuracy, 10 Hz); a Northrop Grumman LITEF LCI-100N tactical IMU (200 Hz, heading accuracy $1\sigma \leq 0.35^\circ$); two Indurad ISDR-C mmWave radars (72–82 GHz, 360°, 1 mm range resolution, 20 Hz); and a ZED2 stereo camera with built-in IMU. Wheel-odometry estimates are published by the Robotnik base controller on `/robot/robotnik_base_control/odom`. All sensors are jointly calibrated; extrinsic transforms are provided as ROS `/tf_static` topics in the bag files.

Data are collected along two routes at 0.15 m/s: an **80 m ring loop** (547 s) with geometry ranging from structured concrete corridors to unstructured rock tunnels, waterlogged sections, and reflective walls; and a **20 m straight tunnel** (134 s) instrumented with 18 spherical control points at 2 m intervals and planar reference targets. The narrowest section is only 0.95 m wide and is the most constrained passage in any published underground SLAM dataset [6]. Table I summarizes key dataset properties compared to representative prior underground benchmarks.

Ground truth is provided by a RIEGL VZ-400i TLS (single-point accuracy 5 mm, declared precision 3 mm), with registra-

tion standard deviations of 0.0019 m (80 m area) and 0.0013 m (20 m area). Control point coordinates in the 20 m area are independently measured with a Trimble S8 total station (1" angular accuracy, 1 mm+1 ppm EDM precision).

IV. METHODOLOGY

A. System Overview

We implement a multi-sensor SLAM pipeline in Python using Open3D for point cloud registration and mapping. The baseline combines wheel-odometry motion priors with LiDAR scan-to-submap registration for pose estimation and global map construction. We extend this framework with millimeter-wave radar as a parallel motion constraint to improve robustness when LiDAR observations become sparse, noisy, or partially occluded. A deliberate LiDAR degradation simulator enables controlled evaluation across eight operating modes spanning LiDAR-only, radar-only, and fused variants under clean and degraded conditions.

B. Sensors and Motion Prior

The system uses a Robosense RS-LiDAR-16 3D LiDAR (~ 10 Hz), two Indurad ISDR-C millimeter-wave radars, and wheel encoder odometry (~ 50 Hz). Sensor-to-base extrinsics apply the dataset-provided translation offsets with identity rotation, since the drivers already correct coordinate axes.

Wheel odometry is temporally interpolated to provide a per-scan relative pose initial estimate. All poses are transformed into a unified frame with the start position as origin and initial heading aligned to the $+X$ axis; heading is unwrapped prior to interpolation to prevent $\pm\pi$ discontinuities.

C. LiDAR Processing and Scan Matching

Each incoming scan is range-filtered (1–50 m) to remove self-returns and unreliable long-range points, voxel-downsampled at 0.15 m, and assigned surface normals via hybrid KD-tree search (0.6 m radius, 30 neighbors) for use in point-to-plane ICP. A sliding-window local submap retains the 20 most recent registered scans merged at 0.25 m voxels.

The odometry-predicted pose initializes ICP against the local submap using a multi-strategy approach: point-to-plane ICP is first attempted at 1.0 m correspondence distance, and if fitness exceeds 0.3 the result is accepted. Otherwise, two fallbacks are attempted (coarse point-to-plane at 3.0 m and point-to-point at 2.0 m) and the highest-fitness candidate is retained. When fewer than 20 points survive (under heavy degradation), the odometry prediction is used directly. After each ICP step, the output rotation is projected to pure yaw and the Z coordinate is attenuated by 0.9 to enforce a near-planar motion assumption consistent with ground-based traversal.

D. Radar Scan Construction and Radar-Only SLAM

For each LiDAR scan at time t , radar measurements from both ISDR sensors within a ± 4 s accumulation window are collected, range-filtered to 0.5–5.0 m, transformed to the base frame, and motion-compensated to pose t using odometry. The merged two-sensor cloud is filtered via statistical outlier

removal ($k = 8, 1.5\sigma$) followed by radius filtering (5 neighbors minimum, 1.0 m radius) to suppress multipath clutter.

In radar-only mode, LiDAR ICP is disabled but LiDAR scans are still used for map construction. Each radar scan is matched against a radar-specific submap (10 most recent scans, 0.15 m voxels) via two-stage point-to-point ICP at 0.5 m and 0.25 m correspondence distances. Results are accepted only when the inlier count exceeds 10 and fitness exceeds 0.2. The accepted radar pose is blended with the odometry prediction using a conservative fitness-dependent weight:

$$w_{\text{radar}} = \min(0.5, 0.6 \cdot f_{\text{radar}}) \quad (1)$$

The 0.5 cap prevents radar from dominating the pose estimate even at high fitness, reflecting radar’s coarser angular resolution over short motion increments.

E. Radar-LiDAR Fusion

The fusion pipeline augments the LiDAR baseline with radar as a parallel motion constraint. LiDAR and radar ICP are executed independently against their respective submaps, and the resulting pose estimates are blended with fixed per-regime weights reflecting expected relative reliability:

$$w_{\text{radar}} = \begin{cases} 0.3 & \text{no degradation} \\ 0.7 & \text{standard degradation} \\ 0.9 & \text{heavy degradation} \end{cases} \quad (2)$$

When a valid radar ICP result is available and the radar submap contains at least 50 points, translation is fused as $\mathbf{t}_{\text{fused}} = (1 - w)\mathbf{t}_{\text{LiDAR}} + w\mathbf{t}_{\text{radar}}$ and heading as $\psi_{\text{fused}} = \psi_{\text{LiDAR}} + w \cdot \Delta\psi$, where $\Delta\psi$ is the shortest-arc yaw difference. Under catastrophic LiDAR failure (<20 points surviving degradation), LiDAR ICP is skipped entirely and radar corrects directly against the odometry prediction using the same blending. This fallback is a key mechanism by which radar fusion recovers trajectory integrity under heavy degradation.

F. LiDAR Degradation Simulation

To evaluate sensor fusion under perceptually degraded conditions, we simulate LiDAR degradation at two severity levels representative of underground sensing hazards such as airborne dust, humidity, and sensor contamination. Degradation is applied to the raw scan before ICP, while the global map is constructed from the unmodified scan; this isolation ensures that observed map-to-ground-truth differences across configurations are attributable to pose estimation error rather than per-scan noise propagating directly into the map.

Three physically-motivated effects are modeled: *stochastic dropout* (random point deletion, simulating absorbed returns), *along-beam range noise* (Gaussian perturbation along each beam direction, simulating multi-path and partial returns), and *spurious returns* (additional points injected at random azimuths and random ranges within 3 m, simulating particulate backscatter).

The *standard* preset (moderate dust or humid air) applies 8% dropout, 0.02 m range noise, and 2% spurious returns. The *heavy* preset (dense dust or contamination) applies 25%

dropout, 0.05 m range noise, and 8% spurious returns. Spurious ratios are computed relative to the count of real points surviving dropout so degradation effects compound realistically.

G. Evaluation Protocol

We evaluate each configuration against the high-precision RIEGL VZ-400i TLS reference point cloud provided with the SubSurfaceGeoRobo dataset. Registered LiDAR scans are fused into a global map using the estimated poses and compared to the RIEGL reference via nearest-neighbor cloud-to-cloud (C2C) distance. Alignment is performed in Open3D: both clouds are centered on their centroids and voxel-downsampled to 0.5 m, FPFH features are computed on each, and a RANSAC feature-matching stage is followed by point-to-plane ICP refinement. This is repeated across five independent RANSAC initializations and the transform with the highest ICP fitness is retained. To ensure fair cross-configuration comparison, the alignment transform is computed once using the clean-baseline map and then applied to every other configuration’s map, so all configurations are evaluated in a shared coordinate frame. For each aligned SLAM cloud, per-point nearest-neighbor distances to the GT cloud are computed via KD-tree search with distances above 5 m gated as outliers. We report Mean, Median, RMSE, and 90th-percentile of the resulting per-point distance distribution.

We evaluate eight configurations in total: LiDAR-only and LiDAR+Radar under each of three degradation levels (clean, standard, heavy), plus Odometry-Only and Radar-Only sensor ablations. This factorial design isolates radar’s contribution at each degradation level while providing single-sensor performance bounds.

V. RESULTS

A. Implementation Details

All experiments are conducted inside a Docker container running Ubuntu 20.04 with ROS Noetic on a host machine with an Intel Core i7 CPU and an NVIDIA RTX 3060 GPU. The SLAM pipeline is implemented in Python 3.8 with Open3D 0.17. Because the SubSurfaceGeoRobo dataset provides a dense reference *scan* rather than a timestamped reference *trajectory*, C2C mean distance and RMSE serve as our primary accuracy metrics; they capture both trajectory drift and map consistency in a single aligned comparison (evaluation protocol detailed in Section IV-G). All results reported are for the **80 m ring sequence** (5,613 processed LiDAR scans, ~547 s).

B. Baseline LiDAR Odometry Results

Table II reports C2C mean distance and RMSE for the LiDAR-odometry pipeline and the pure wheel-odometry fallback, compared against prior methods evaluated on the same 80 m ring test field by Li et al. [7].

The LiDAR-odometry pipeline reduces C2C mean distance from 0.449 m (wheel odometry alone) to 0.311 m, confirming

TABLE II: Map Accuracy on the 80 m Ring Sequence

Method	C2C Mean (m)	C2C RMSE (m)
Odom only (no SLAM)	0.449	0.698
Ours (LiDAR baseline)	0.311	0.616
HDL Graph SLAM [7]	0.123	—
RTAB-Map [7]	0.370	—

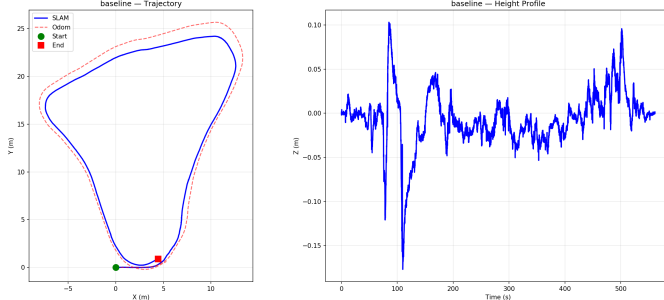


Fig. 1: Baseline LiDAR-odometry trajectory (blue) vs. wheel-odometry estimate (dashed pink) for the 80 m ring sequence (bird’s-eye view, left) and height profile over time (right). Start: green circle; end: red square.

that ICP-based scan-to-submap registration provides a substantial correction over the raw encoder estimate. Our result of 0.311 m falls between HDL Graph SLAM (0.123 m) and RTAB-Map (0.370 m) evaluated on the same 80 m ring test field by Li et al. [7]; our pipeline outperforms RTAB-Map without any graph-optimization back-end or loop closure, and trails HDL Graph SLAM by 0.188 m in mean C2C, a gap attributable to the absence of a full pose-graph smoother and loop-closure detection module in our lightweight design.

The ICP fitness score (inlier ratio) averages ~ 0.99 across all 5613 scans, with the fitness score dropping below 0.15 (the odometry-fallback threshold) in well under 1% of frames. This indicates that the sliding-window local submap provides sufficient geometric context for robust scan matching throughout the 80 m traverse under clean LiDAR conditions.

Fig. 1 shows the estimated trajectory and height profile for the baseline run. The 2-D trajectory closely follows the ring geometry; residual height excursions remain within ± 0.15 m thanks to the ground-plane constraint, confirming that Z-axis drift is effectively suppressed.

C. Radar-Augmented Results

Table III compares C2C accuracy across all six pipeline configurations: the three degradation levels (none, standard, heavy) each run with and without radar fusion.

Radar constraints provide their largest benefit in the narrow corridor sections (approximately 30–50 m into the 80 m ring) where the two-wall geometry creates longitudinal degeneracy in LiDAR scan-matching. In these segments, radar constraints provide the most meaningful correction as LiDAR scan-matching fails along the tunnel axis. In the wider, geometrically richer junction areas, LiDAR registration dominates and

TABLE III: Map Accuracy vs. LiDAR Degradation Level (80 m Ring)

Configuration	C2C Mean (m)	C2C RMSE (m)	90th pct (m)
Baseline (no degr.)	0.311	0.616	0.423
Baseline + Radar ($w = 0.3$)	0.316	0.615	0.420
Standard degraded	0.631	1.037	1.717
Std. degraded + Radar ($w = 0.7$)	0.404	0.672	0.692
Heavy degraded	1.015	1.524	2.768
Heavy degraded + Radar ($w = 0.9$)	0.394	0.656	0.680

radar adds minimal improvement, as radar ICP is similarly degenerate in open spaces with few stable landmark clusters.

D. Robustness Under LiDAR Degradation

We report C2C accuracy under each simulated degradation level below.

1) *Standard degradation.*: The standard-degradation preset (8% dropout, 0.02 m noise, 2% spurious) degrades mean C2C from 0.311 m to 0.631 m. Radar fusion at $w = 0.7$ recovers accuracy to 0.404 m, a **36% reduction in mean C2C** and a **35% reduction in RMSE** relative to the degraded LiDAR baseline. The 90th-percentile distance drops from 1.717 m to 0.692 m, indicating that the worst-case registration errors are cut by more than half.

2) *Heavy degradation.*: The heavy preset (25% dropout, 0.05 m noise, 8% spurious) causes mean C2C to reach 1.015 m and RMSE 1.524 m. Severe trajectory corruption is visible as a compressed and deformed ring shape in Fig. 2. Radar fusion at $w = 0.9$ reduces mean C2C to 0.394 m and RMSE to 0.656 m, a **61% and 57% reduction** respectively. Notably, the fused system under heavy degradation (0.394 m) achieves accuracy comparable to the clean LiDAR baseline (0.311 m), and outperforms the standard-degraded LiDAR-only result (0.631 m). The 90th-percentile distance collapses from 2.768 m to 0.680 m, recovering from near-catastrophic map corruption to a quality approaching the undegraded case.

Fig. 3 summarizes the C2C metrics across all configurations as a grouped bar chart.

E. Radar-only SLAM

The radar-only configuration achieves a mean C2C distance of 0.579 m, median of 0.354 m, RMSE of 0.876 m, and 90th-percentile error of 1.277 m. Across all metrics this exceeds the wheel-odometry-only baseline (mean 0.449 m, RMSE 0.699 m), indicating that radar ICP in isolation introduces additional pose noise rather than reducing drift.

The primary limitation is the sparse, two-dimensional nature of the Indurad sensor: the lack of vertical structure and low point density in narrow tunnel cross-sections produce ambiguous point-to-point correspondences, and the fitness-dependent weight $w_{\text{radar}} = \min(0.5, 0.6 \cdot f_{\text{radar}})$ frequently saturates near its lower bound. As shown in Fig. 4, radar ICP does correct cross-track drift relative to raw odometry, confirming that the sensor carries useful lateral constraint; the degradation relative to odometry arises along the tunnel axis,

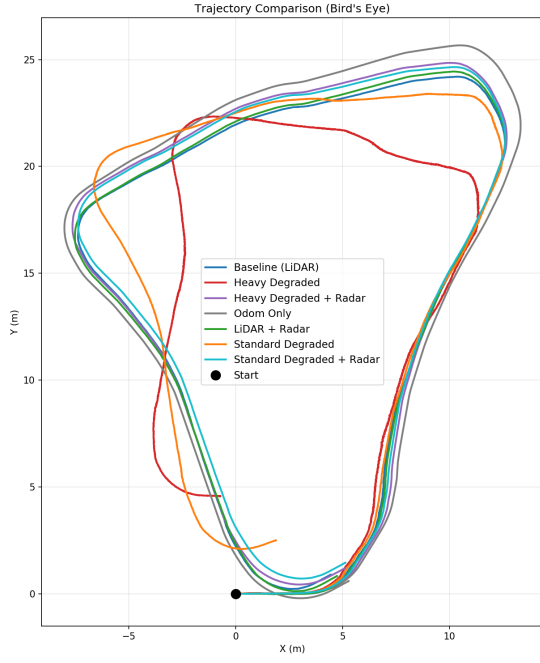


Fig. 2: Bird’s-eye-view trajectory comparison for all six configurations. The heavy-degraded LiDAR-only trajectory (red) is visibly deformed relative to the baseline ring shape; adding radar (purple) restores the loop geometry.

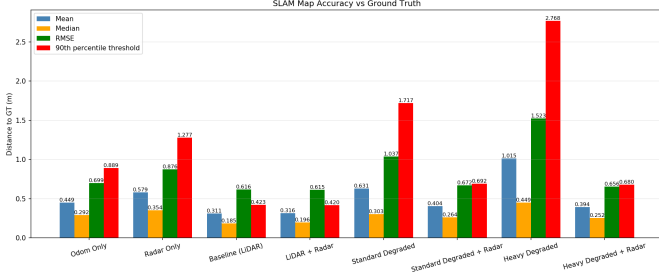


Fig. 3: C2C distance to RIEGL ground truth for all configurations. Mean (blue), median (orange), RMSE (green), and 90th-percentile (red) distances are shown. The paired columns illustrate the consistent benefit of radar fusion, especially under heavy degradation.

where sparse azimuthal returns cannot resolve longitudinal slip. These results establish radar as a viable *fusion* signal rather than a standalone localizer, motivating the fixed-weight blending evaluated in Sections V-C and V-D.

F. Qualitative Map Analysis

Fig. 5 shows side-by-side views of the point cloud maps produced by the LiDAR-odometry baseline and the pure wheel-odometry fallback, both aligned to the RIEGL TLS ground truth using the FPFH+RANSAC+ICP pipeline described in Section IV-G.

In the baseline map, the highest C2C distances are concentrated in two regions: the narrow 0.95 m section near the shaft

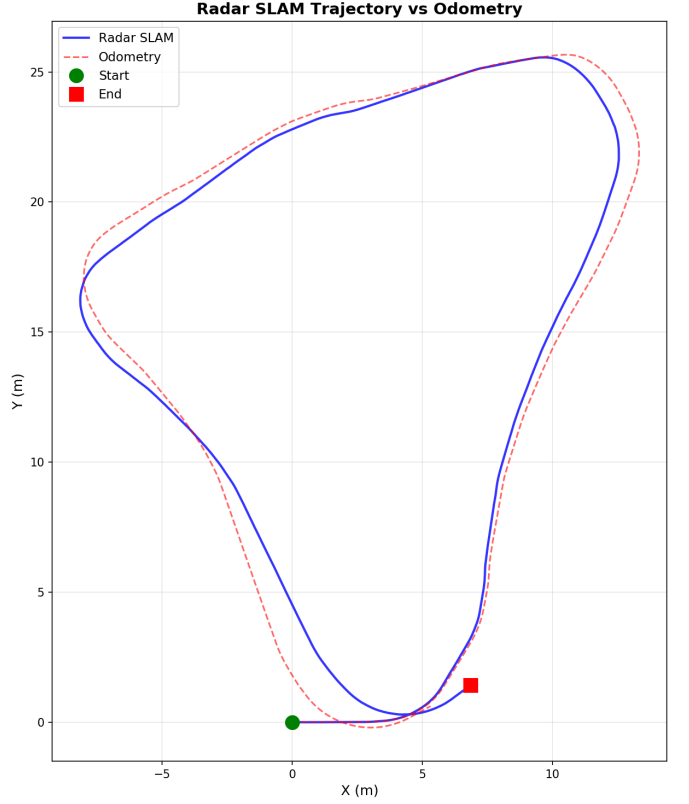


Fig. 4: Radar-only SLAM trajectory (blue) vs. wheel-odometry estimate (dashed pink) for the 80 m ring sequence. The radar ICP successfully corrects the odometry drift along cross-track directions, though the sparser point density and 2-D nature of the Indurad sensors limit absolute accuracy relative to the LiDAR baseline.

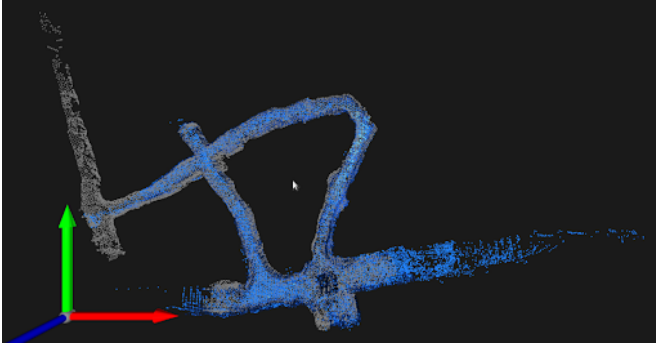
entrance, and a long straight corridor segment approximately 30–50 m along the ring where the tunnel geometry is featureless for several consecutive scans. The radar-augmented map shows reduced error magnitude in the straight corridor region, while the narrow shaft entrance section remains challenging for both pipelines due to sensor occlusion and the physical limits of the RS-LiDAR-16 field of view in such confined geometry.

VI. DISCUSSION

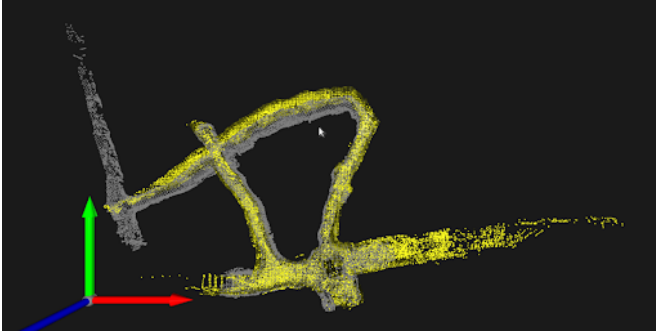
A. Radar Integration Effectiveness and Limitations

The radars produce sparse 2-D point clouds with limited geometric detail relative to the 3-D LiDAR. Our results show that radar fusion provides consistent accuracy improvements under degraded conditions (36% mean error reduction under standard degradation and 61% under heavy degradation) while introducing negligible effects under clean conditions. This pattern indicates that radar is most valuable when LiDAR registration weakens, rather than a general improvement.

The radar-only result (0.579 m) performing worse than odometry alone (0.449 m) indicates that radar ICP in isolation introduces pose noise, particularly along the tunnel axis where



(a) 3D map comparison between baseline LiDAR SLAM (blue) and ground truth (grey)



(b) 3D map comparison between Odom only mapping (yellow) and ground truth (grey)

Fig. 5: Side-by-side Point Cloud Maps

sparse radar points around the tunnel cross-section cannot resolve longitudinal motion. The fitness-dependent weighting mitigates this in fusion, but the radar's contribution is likely uneven across the route, where transitional zones where corridor geometry changes abruptly would be expected to provide better radar landmarks than long unobstructed straight sections.

Additionally, the 2-D nature of the Indurad sensors means no elevation constraint is contributed by radar, leaving Z-axis drift entirely dependent on ground-plane enforcement. A 4-D mmWave radar providing elevation measurements would address this limitation directly.

B. Implications for Geomonitoring

The SubSurfaceGeoRobo dataset targets SLAM-based underground geomonitoring, where accurate volumetric reconstruction is needed to detect structural deformation over repeated survey campaigns. Our baseline achieves 0.311 m mean C2C distance, which is sufficient for coarse structural mapping but falls short of the centimeter-level precision needed to detect small deformation events. However, our results suggest that radar fusion is particularly relevant in this context: under degraded conditions representative of active mining environments radar fusion recovers accuracy to 0.394 m even under heavy degradation, keeping the system operational where LiDAR alone fails entirely. For geomonitoring applications, maintaining any reliable map under degraded conditions is

more valuable than achieving peak accuracy only under ideal ones. A practical workflow would combine our SLAM pipeline for rapid trajectory estimation with a post-processing cloud-to-cloud alignment step against a prior TLS reference scan to achieve the fine registration needed for deformation detection.

VII. CONCLUSION

We presented a benchmarking study of LiDAR-odometry and LiDAR-radar fusion SLAM on the SubSurfaceGeoRobo underground mine dataset, evaluating eight configurations spanning LiDAR-only, radar-only, and fused variants under clean and simulated degradation conditions.

Under clean conditions, our LiDAR baseline achieves 0.311 m mean C2C distance, outperforming RTAB-Map (0.370 m) without any graph-optimization back-end. Simulated LiDAR degradation significantly degrades localization accuracy. Standard degradation doubles mean error to 0.631 m, while heavy degradation reaches 1.015 m with visible trajectory deformation. Radar fusion reduces mean error by 36% under standard degradation and 61% under heavy degradation, with the fused heavy-degraded result (0.394 m) approaching the clean baseline. Under non-degraded conditions, radar fusion introduces negligible differences (0.316 m vs. 0.311 m), confirming that radar integration does not measurably impact performance when LiDAR is healthy.

Radar proves most valuable in geometrically transitional corridor sections where LiDAR scan-matching becomes rank-deficient, demonstrating the complementary nature of the two sensing modalities in challenging subterranean environments.

Future work can explore: (i) adding loop closure to close the accuracy gap with state-of-the-art methods; (ii) replacing the 2-D Indurad radars with a 4-D mmWave sensor to recover elevation constraints; and (iii) testing on additional underground sequences to validate generalization.

Code and Video. The full implementation and evaluation scripts are publicly available at:

[Github Link](#)

A demonstration video summarizing the project approach and results is available at:

<https://www.youtube.com/watch?v=CVoWS0OxvvE>

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